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**Course: DSA0602-Data handling and
visuilization**

EXPERIMENT 1: Monthly Sales Analysis

```
library(ggplot2)  
library(plotly)
```

```
sales <- data.frame(  
  Month = c("Jan", "Feb", "Mar", "Apr", "May"),  
  Sales = c(1200, 1500, 1800, 1600, 2000)  
)
```

```
print(sales)
```

```
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Monthly Sales Trend",  
    x = "Month",  
    y = "Sales"  
  ) +  
  theme_minimal()
```

```
ggplot(sales, aes(x = Month, y = Sales, fill = Month)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Monthly Sales Comparison",  
    x = "Month",  
    y = "Sales"  
  ) +  
  theme_minimal()
```

```
ggplot(sales, aes(x = Month, y = Sales)) +  
  geom_point(size = 4, color = "blue") +  
  geom_smooth(aes(group = 1), method = "lm", se = FALSE, color = "red") +  
  labs(  
    title = "Monthly Sales Scatter Plot",  
    x = "Month",  
    y = "Sales"  
  ) +  
  theme_minimal()
```

```
p1 <- plot_ly(  
  sales,  
  x = ~Month,
```

```

y = ~Sales,
type = "scatter",
mode = "lines+markers"
)

```

```

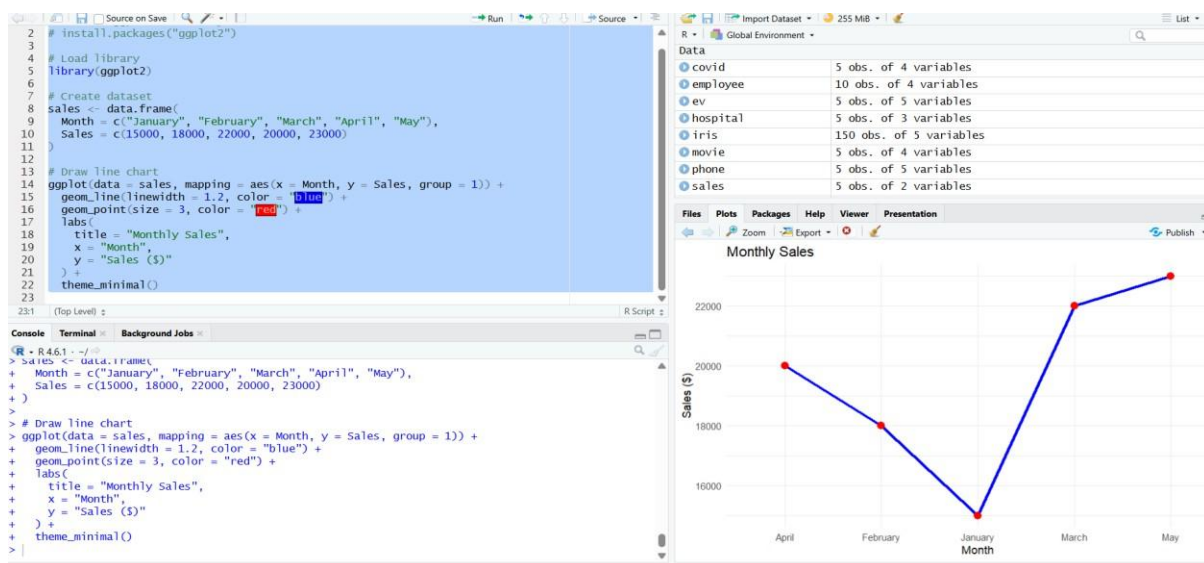
p2 <- plot_ly(
  sales,
  x = ~Month,
  y = ~Sales,
  type = "bar"
)

```

```

subplot(p1, p2, nrows = 2)

```



EXPERIMENT 2: Customer Feedback Analysis

PROGRAM

```

library(ggplot2)
library(wordcloud)
library(RColorBrewer)

```

```

customer <- data.frame(
  Age = c(20, 25, 30, 35, 40),
  Satisfaction = c("High", "Medium", "High", "Low", "Medium")
)

```

```

print(customer)

```

```

ggplot(customer, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Customer Age Distribution",
    x = "Age",

```

```

    y = "Frequency"
  ) +
  theme_minimal()

```

```

pie(
  table(customer$Satisfaction),
  col = rainbow(3),
  main = "Customer Satisfaction Level"
)

```

```

customer$Age_Group <- c("20-25", "20-25", "30-35", "30-35", "40+")

```

```

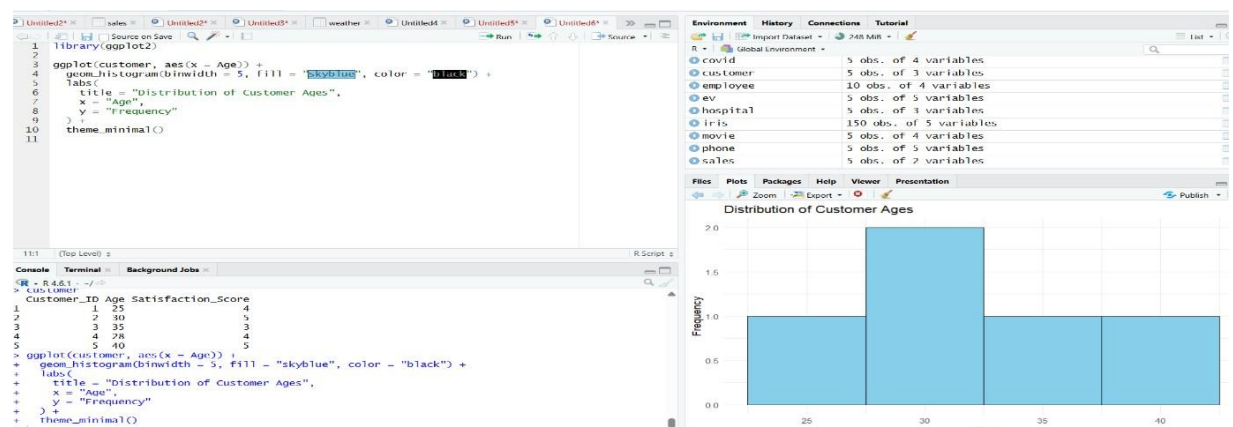
ggplot(customer, aes(x = Age_Group, fill = Satisfaction)) +
  geom_bar() +
  labs(
    title = "Satisfaction by Age Group",
    x = "Age Group",
    y = "Count"
  ) +
  theme_minimal()

```

```

wordcloud(
  words = customer$Satisfaction,
  freq = c(2, 2, 1, 1, 2),
  colors = brewer.pal(8, "Dark2")
)

```



EXPERIMENT 3: Employee Performance Analysis

```

library(ggplot2)

```

```

library(plotly)

```

```

employee <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales",

```

```
"HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)
```

```
print(employee)
```

```
ggplot(employee, aes(x = Years_of_Service, y =  
Performance_Score, group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Employee Performance Trend",  
    x = "Years of Service",  
    y = "Performance Score"  
  ) +  
  theme_minimal()
```

```
ggplot(employee, aes(x = Department, fill = Department))  
+  
  geom_bar() +  
  labs(  
    title = "Employees by Department",  
    x = "Department",  
    y = "Number of Employees"  
  ) +  
  theme_minimal()
```

```
ggplot(employee, aes(x = Years_of_Service, y =  
Performance_Score)) +  
  geom_point(size = 4, color = "blue") +  
  geom_smooth(method = "lm", se = FALSE, color =  
"red") +  
  labs(  
    title = "Years of Service vs Performance",
```

```

x = "Years of Service",
y = "Performance Score"
) +
theme_minimal()

```

```

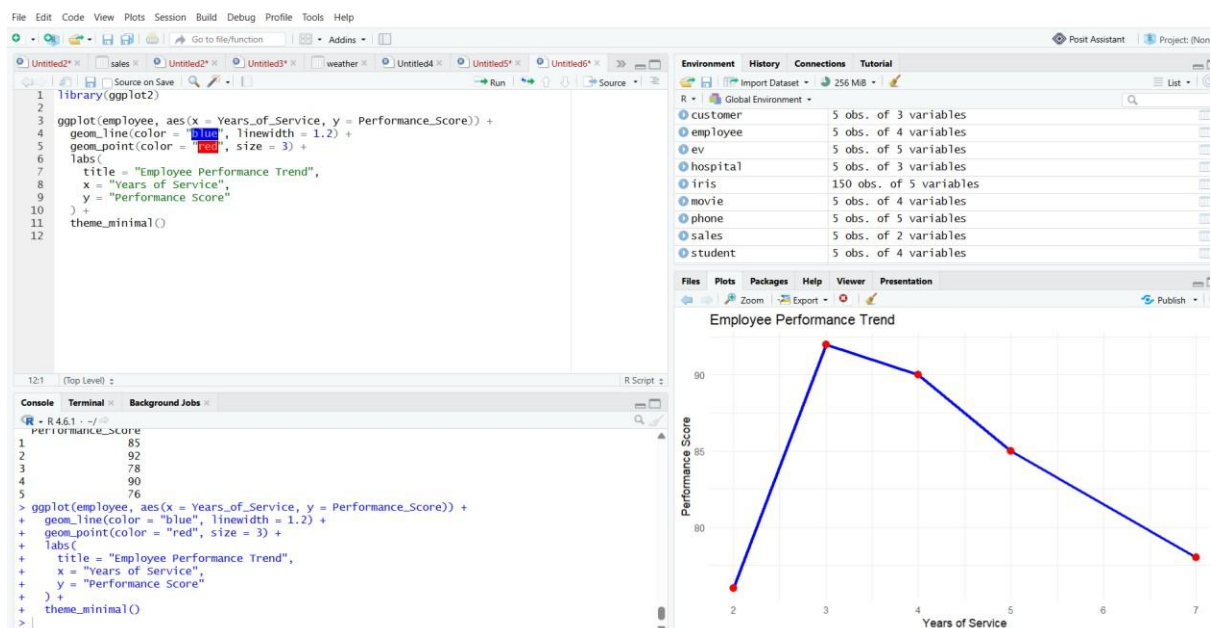
p1 <- plot_ly(
  employee,
  x = ~Years_of_Service,
  y = ~Performance_Score,
  type = "scatter",
  mode = "lines+markers"
)

```

```

p2 <- plot_ly(
  employee,
  x = ~Department,
  type = "bar"
)

```



EXPERIMENT 4: Product Inventory Analysis

PROGRAM

```
library(ggplot2)
library(plotly)

inventory <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture"),
  Price = c(500, 650, 450, 700, 550),
  Quantity = c(250, 175, 300, 200, 220)
)

print(inventory)

ggplot(inventory, aes(x = Product_Name, y = Quantity, fill = Product_Name)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory",
    x = "Product Name",
    y = "Quantity"
  ) +
  theme_minimal()

ggplot(inventory, aes(x = Category, y = Quantity, fill = Product_Name)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Inventory by Category",
    x = "Category",
    y = "Quantity"
  ) +
  theme_minimal()

ggplot(inventory, aes(x = Price, y = Quantity)) +
  geom_point(size = 4, color = "blue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Price vs Quantity",
    x = "Price",
    y = "Quantity"
  ) +
  theme_minimal()

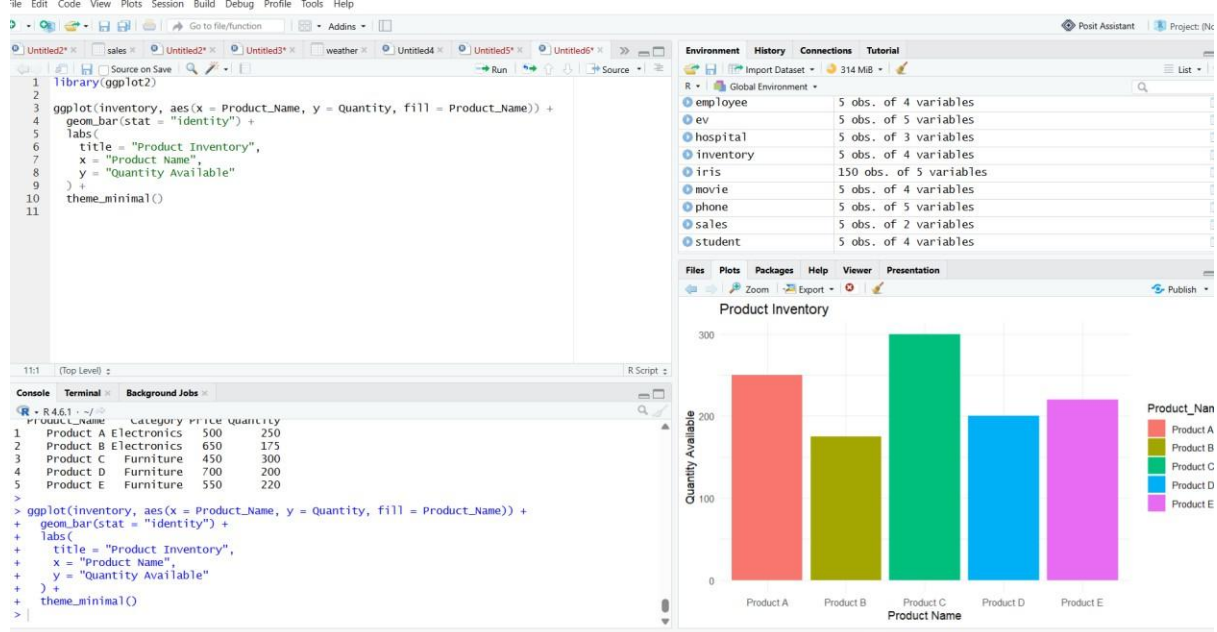
p1 <- plot_ly(
  inventory,
  x = ~Product_Name,
  y = ~Quantity,
  type = "bar"
)

p2 <- plot_ly(
  inventory,
  x = ~Price,
  y = ~Quantity,
```

```
type = "scatter",  
mode = "markers"
```

```
)
```

```
subplot(p1, p2, nrow = 2)
```



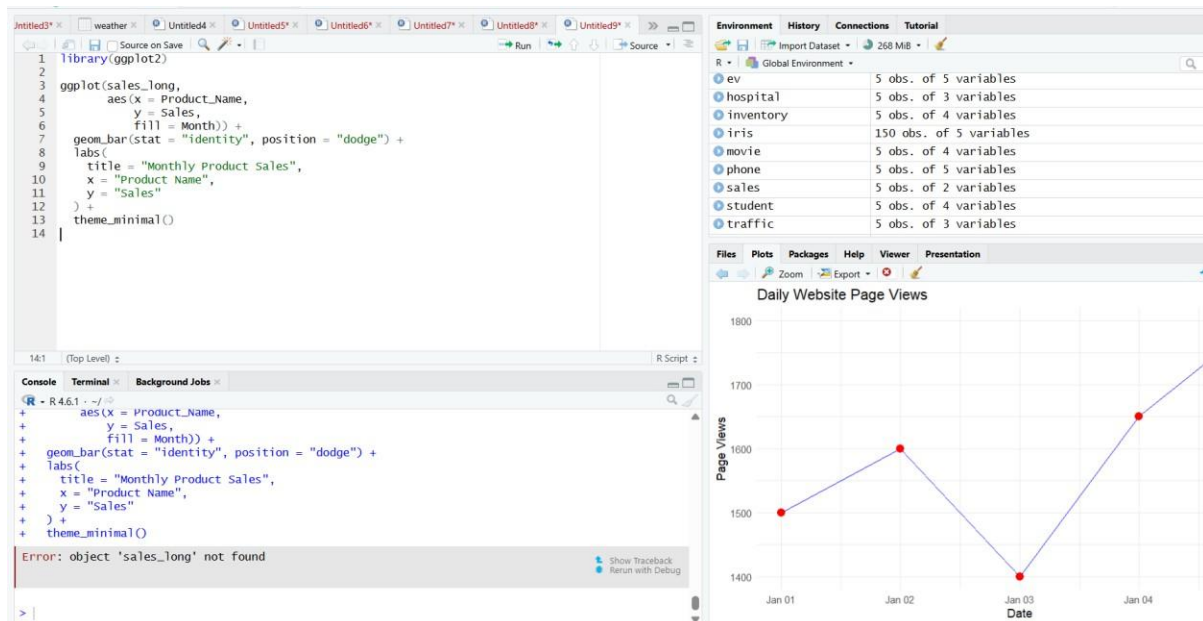
SET 5: Website Analytics

```
library(ggplot2)
library(plotly)
library(tidyr)

traffic <- data.frame(
  Date = as.Date(c(
    "2023-01-01",
    "2023-01-02",
    "2023-01-03",
    "2023-01-04",
    "2023-01-05"
  )),
  Page_Views = c(
    1500,
    1600,
    1400,
    1650,
    1800
  ),
  CTR = c(
    2.3,
    2.7,
    2.0,
    2.4,
    2.6
  )
)

print(traffic)

ggplot(traffic, aes(x = Date, y = Page_Views, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Website Page Views",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()
```

Set 6-Product Sales Analysis

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
library(ggplot2)
```

```
library(tidyr)
```

```
sales <- data.frame(
```

```
  Product_ID = c(1, 2, 3),
```

```
  Product_Name = c("Product A", "Product B", "Product C"),
```

```
  January = c(2000, 1500, 1200),
```

```
  February = c(2200, 1800, 1400),
```

```
  March = c(2400, 1600, 1100)
```

```
)
```

```
sales_long <- pivot_longer(
```

```
  sales,
```

```
  cols = c(January, February, March),
```

```
  names_to = "Month",
```

```
  values_to = "Sales"
```

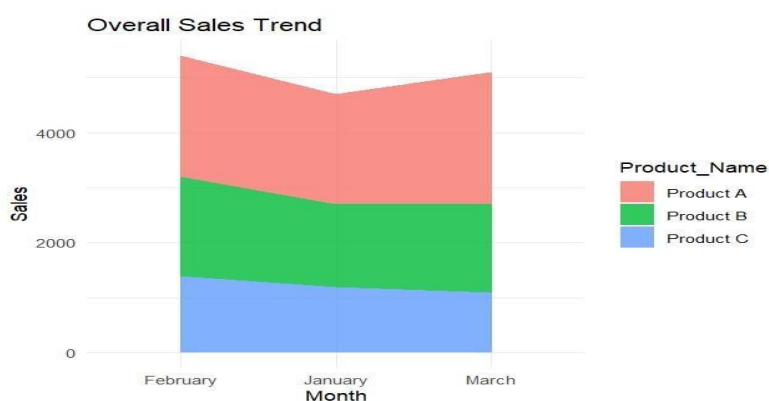
```
)
```

```

ggplot(sales_long, aes(x = Product_Name, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "First Quarter Product Sales",
    x = "Product",
    y = "Sales"
  ) +
  theme_minimal()

```

Output:



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
library(ggplot2)
```

```
library(tidyr)
```

```

sales <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

```

```
sales_long <- pivot_longer(  
  sales,  
  cols = c(January, February, March),  
  names_to = "Month",  
  values_to = "Sales"  
)
```

```
sales_long$Month <- factor(  
  sales_long$Month,  
  levels = c("January", "February", "March")  
)
```

```
ggplot(sales_long, aes(x = Month, y = Sales, fill = Product_Name, group =  
Product_Name)) +  
  geom_area(alpha = 0.8) +  
  labs(  
    title = "Overall Sales Trend",  
    x = "Month",  
    y = "Sales"  
  ) +  
  theme_minimal()
```

Output:



3. Build a table to show the monthly sales data for each product. Label the table elements.

```
sales <- data.frame(
```

```
  Product_ID = c(1,  
2, 3),
```

```
  Product_Name =  
c("Product A",  
"Product B",  
"Product C"),
```

```
  January_Sales =  
c(2000, 1500, 1200),
```

```
  February_Sales =  
c(2200, 1800, 1400),
```

```
  March_Sales =  
c(2400, 1600, 1100)
```

```
)
```

```
print(sales))
```

Output:

```
Product_Name January February March
```

1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100